

ASSIGNMENT #07

Course: CHE 341: CHEMICAL PROCESS CONTROL
Term: Spring 2026
Professor: Dr. Michael Caracotsios
Office: EIB 244

Cell Phone: (630) 696-0934
E-mail: mcaracot@uic.edu

Name : _____ **Anna Cai** _____

Due Date: Tuesday, May 5th in Class

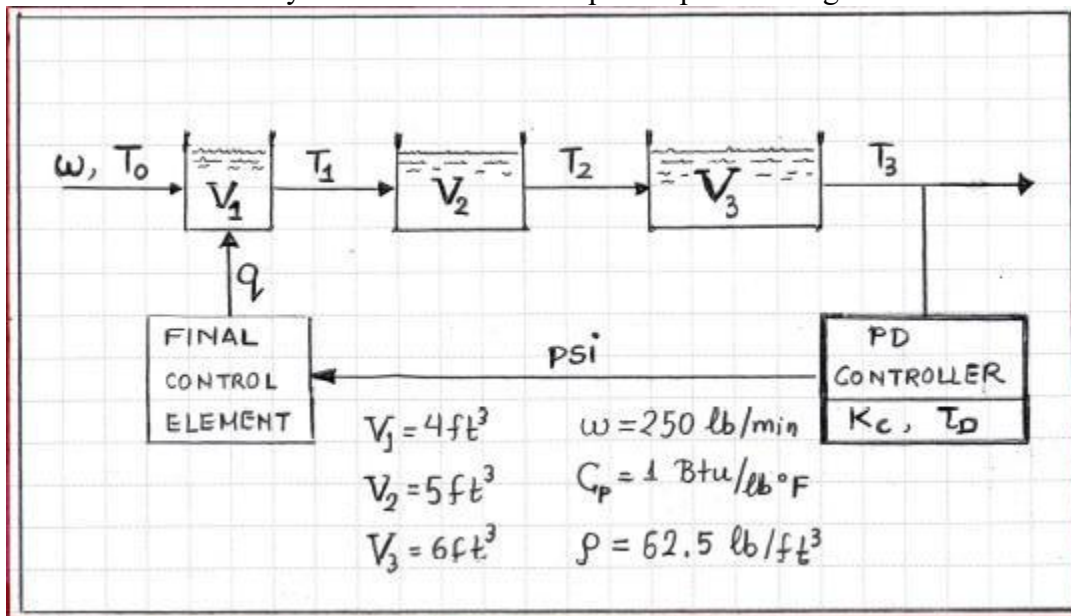
INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
 - Include this cover page with your name in the report.
 - Upload your report as a PDF file by the Due Date and Time(**6:00PM**) on the UIC Blackboard.
 - Place a hard copy of your report as a PDF file by the Due Date and Time(**Closing of the Business Day**) in the TA's mailbox.

 - Assignments that are not typed will not be accepted.
 - Late Assignments will not be accepted without prior approval.
 - You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report
-

Problem 1: Feedback Control of a Thermal System

Consider the thermal system shown in the simplified process diagram below:



The thermal system is currently being controlled by a Proportional Derivative (PD) controller

with gain and rate equal to:

$$K_C = 1500 \frac{\text{Btu/min}}{\text{F}} \quad \tau_D = 0.5 \text{ min}$$

The temperature T_o of

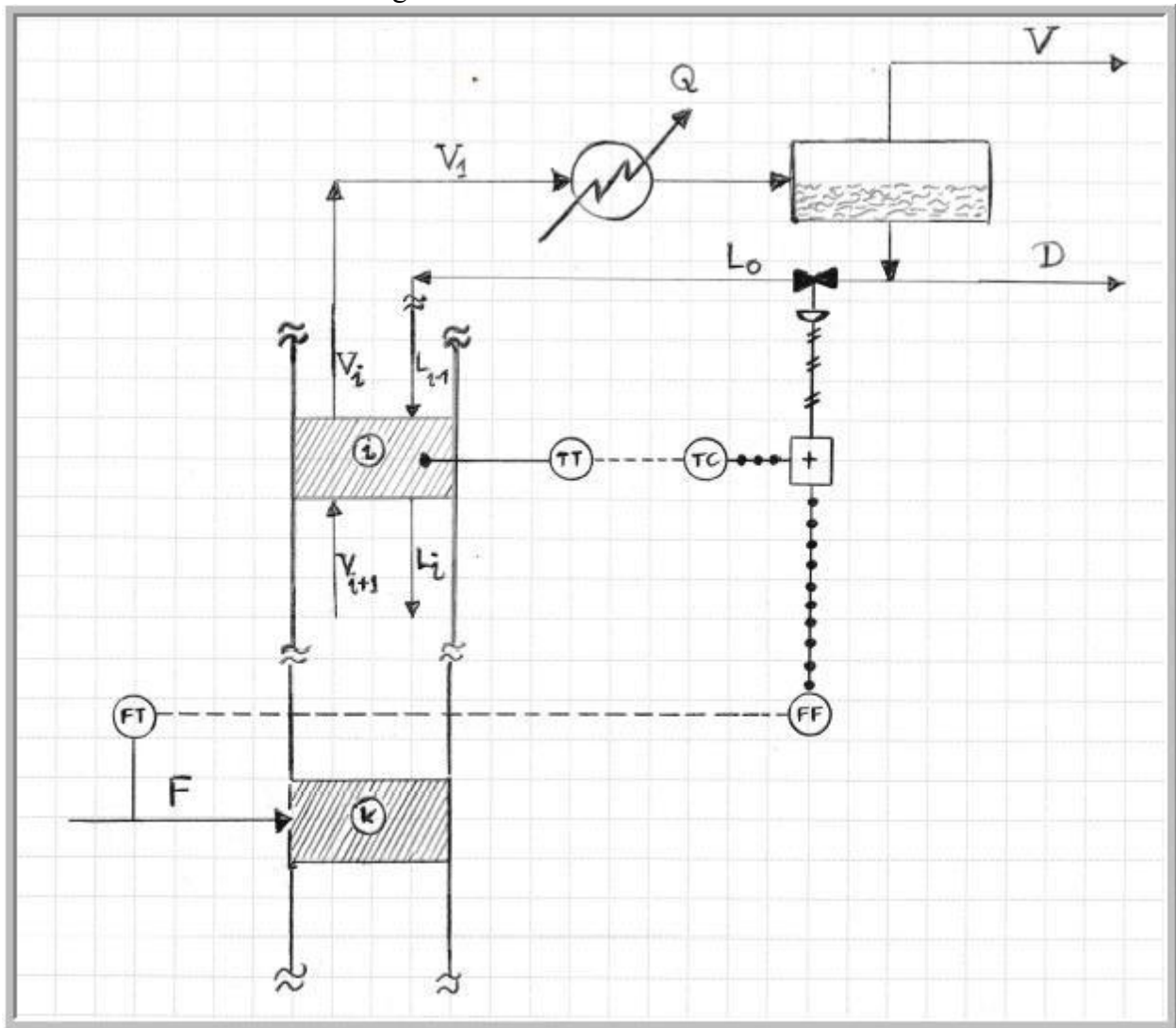
stream is considered a

disturbance.

1. Draw a block diagram of the control system with the transfer functions in each block. Each transfer function must only have the numerical values of its parameters.
2. From this block diagram, determine the closed loop transfer function relating the temperature in tank #3 to a change in the set point.
3. From this block diagram determine the closed loop transfer function relating the temperature in tank #3 to a change in the inlet temperature T_o .
4. Calculate the offset for a unit step change in the inlet temperature T_o .
5. Perform a simulation using the **CHE Calculator.xlsm** Excel tool to verify your theoretical answer in question #4.
6. You have been asked to eliminate the observed offset. Design a controller and calculate its parameters that would accomplish just that. Perform a simulation again, using the **CHE Calculator.xlsm** Excel tool to verify that the observed offset has been eliminated.
7. Calculate the settling time and present it along with the controller parameters in a Table included in a separate page in your final report. Your final report should also include the graphical results for questions #5 and #6.

Problem 2: Feedforward Control of a Distillation Column Tray Temperature

The temperature of a critical tray in a fractionator will be controlled by employing a Feedforward controller as indicated in the diagram shown below.



Using the appropriate step tests the following individual transfer functions models were obtained for the response of the tray temperature to (1) feed changes and (2) reflux flow rate changes:

G_D	$\frac{T_s(\cdot) 20e^{-15s}}{F_s(\cdot) 301s+}$
G_P	$\frac{T_s(\cdot) - 5e^{-2s}}{L_{s_o}(\cdot) 61s+}$

(Equation 1)

- a) Design a **Feedforward** controller to achieve perfect disturbance rejection and show in a control block diagram how it would be implemented. Is the **Feedforward** controller realizable?

Consider the situation in which the model is inaccurate with the actual model process given by:

$$T(s) = \frac{-4e^{-2s}}{6s+1} L(s) + \frac{25e^{-15s}}{30s+1} F(s) \quad (\text{Equation 2})$$

- b) Using the **Feedforward** controller designed in part (a) simulate and plot the response of the temperature under a **Feedforward** control to a step change of $DS=2$ in the feed flow rate
- c) Design a **Feedback** controller for this process using direct synthesis with a $\lambda = 0.25$. In the design, use the parameters of the model assumed to be correct and given in Equation (1). Apply the **Feedback** controller to the actual process without using the **Feedforward** controller and simulate the closed loop response to a step change of $DS=2$ in the feed rate (disturbance)
- d) No apply to the actual process (Equation 2) a **Feedforward-Feedback** control scheme using the **Feedforward** controller designed in (a) and the **Feedback** controller designed in (c). Simulate the closed loop response to a step change of $DS=2$ in the feed rate (disturbance)
- e) Show all three response curves in a single graph and verify whether the **Feedforward** controller performs as expected.

Problem 1: Feedback Control of a Thermal System

1. Block Diagram and Control Variables

The process is a three-tank thermal system with liquid flowing through three well-mixed tanks in series. The controlled variable is the outlet temperature from the third tank, so the process variable is

$$PV = T_3$$

The inlet temperature, T_0 , is treated as the disturbance variable because changes in the inlet stream temperature affect the outlet temperature even when the controller output is unchanged:

$$DV = T_0$$

The manipulated variable is the heat input, q , added to Tank 1 through the final control element. In the feedback loop, the controller compares the set point with the measured value of T_3 and adjusts q to reduce the error.

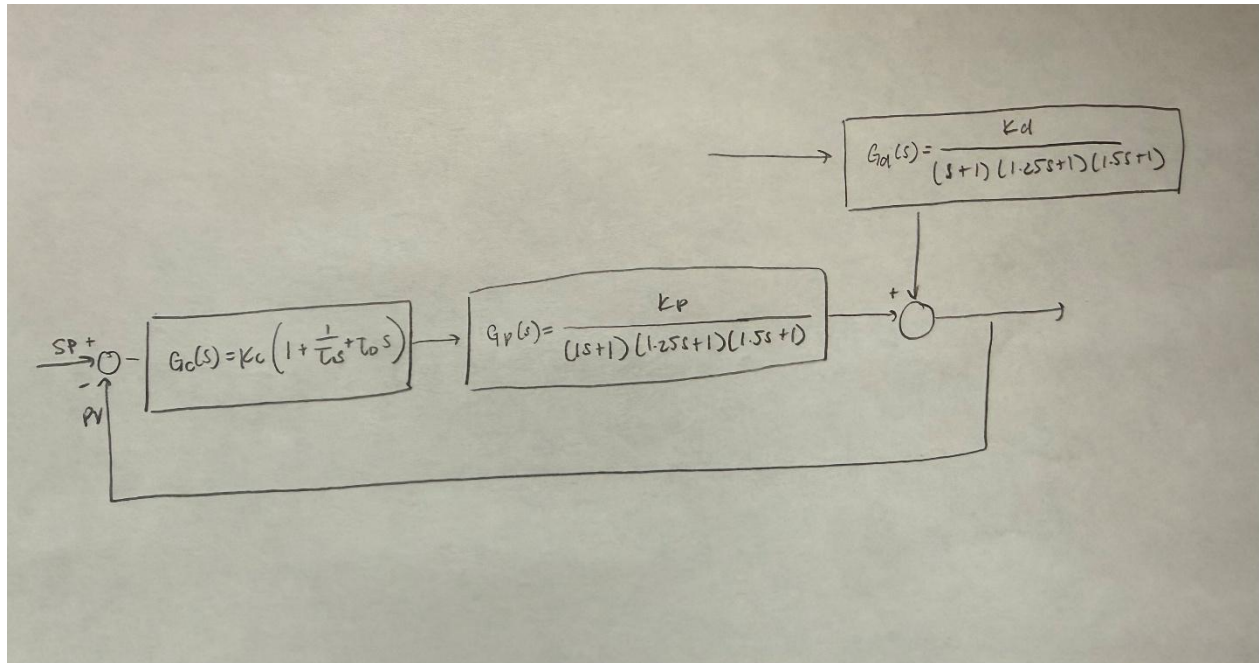


Figure 1: Feedback block diagram for the three-tank thermal system.

The transfer functions used in the block diagram are written below. For the original controller, the integral term is absent because the controller is proportional-derivative only:

$$G_c(s) = 1500(1 + 0.5s)$$

$$G_p(s) = \frac{0.004}{(1.0s + 1)(1.25s + 1)(1.5s + 1)}$$

$$G_d(s) = \frac{1}{(1.0s + 1)(1.25s + 1)(1.5s + 1)}$$

In the CHE Calculator simulation, the three tank lags were entered using the available process, valve, and sensor lag fields. This was done only as a convenient way to preserve the full third-order model. It does not mean that Tanks 2 and 3 are physically valves or sensors.

2. Process Transfer Function for Set-Point Changes

For Tank 1, an unsteady-state energy balance gives

$$\rho V_1 C_p \frac{dT_1}{dt} = \omega C_p (T_0 - T_1) + q$$

To obtain the process transfer function, the inlet temperature disturbance is held constant and only the response to heat input is considered. In deviation-variable form, this becomes

$$\rho V_1 C_p \frac{dT_1}{dt} = -\omega C_p T_1 + q$$

Taking the Laplace transform and solving for $T_1(s)/Q(s)$ gives

$$\frac{T_1(s)}{Q(s)} = \frac{1}{\omega C_p} \frac{1}{\left(\frac{\rho V_1}{\omega}\right)s + 1}$$

The time constant for each tank is

$$\tau = \frac{\rho V}{\omega}$$

Using $\rho = 62.5 \text{ lb/ft}^3$ and $\omega = 250 \text{ lb/min}$:

$$\tau_1 = \frac{(62.5)(4)}{250} = 1.0 \text{ min}$$

$$\tau_2 = \frac{(62.5)(5)}{250} = 1.25 \text{ min}$$

$$\tau_3 = \frac{(62.5)(6)}{250} = 1.5 \text{ min}$$

The process gain for heat input is

$$K_p = \frac{1}{\omega C_p} = \frac{1}{(250)(1)} = 0.004$$

Tanks 2 and 3 do not add heat. They only contribute additional first-order lag behavior, so the complete process transfer function is

$$G_p(s) = \frac{0.004}{(1.0s + 1)(1.25s + 1)(1.5s + 1)}$$

The closed-loop transfer function from set point to T_3 is

$$\frac{PV(s)}{SP(s)} = \frac{G_c(s)G_p(s)}{1 + G_c(s)G_p(s)}$$

Substituting the original PD controller gives

$$\begin{aligned}\frac{PV(s)}{SP(s)} &= \frac{6(1 + 0.5s)}{(1.0s + 1)(1.25s + 1)(1.5s + 1) + 6(1 + 0.5s)} \\ \frac{PV(s)}{SP(s)} &= \frac{3s + 6}{1.875s^3 + 4.625s^2 + 6.75s + 7}\end{aligned}$$

3. Disturbance Transfer Function

The disturbance transfer function describes how a change in inlet temperature T_0 affects T_3 . Since the inlet temperature enters through the feed stream, the disturbance passes through all three tanks in series. The first tank balance for a disturbance change is

$$\rho V_1 C_p \frac{dT_1}{dt} = \omega C_p (T_0 - T_1)$$

Taking the Laplace transform and solving for $T_1(s)/T_0(s)$ gives

$$\frac{T_1(s)}{T_0(s)} = \frac{1}{\left(\frac{\rho V_1}{\omega}\right)s + 1}$$

A sustained inlet temperature change eventually passes completely through each tank, so the steady-state disturbance gain is 1. Therefore,

$$G_d(s) = \frac{1}{(1.0s + 1)(1.25s + 1)(1.5s + 1)}$$

For a disturbance input with the set point held constant, the closed-loop transfer function is

$$\frac{PV(s)}{DV(s)} = \frac{G_d(s)}{1 + G_c(s)G_p(s)}$$

Substituting the original PD controller gives

$$\begin{aligned}\frac{PV(s)}{DV(s)} &= \frac{1}{(1.0s + 1)(1.25s + 1)(1.5s + 1) + 6(1 + 0.5s)} \\ \frac{PV(s)}{DV(s)} &= \frac{1}{1.875s^3 + 4.625s^2 + 6.75s + 7}\end{aligned}$$

4. Offset for a Unit Step Change in T_0

The offset was calculated using the final value theorem. For a unit step change in inlet temperature,

$$DV(s) = \frac{1}{s}$$

The steady-state process-variable deviation is

$$PV_{ss} = \lim_{s \rightarrow 0} sPV(s) = \frac{G_d(0)}{1 + G_c(0)G_p(0)}$$

At steady state, the derivative portion of the controller is zero, so

$$G_c(0) = 1500, \quad G_p(0) = 0.004, \quad G_d(0) = 1$$

$$PV_{ss} = \frac{1}{1 + (1500)(0.004)} = \frac{1}{7} = 0.1429$$

Therefore, the offset for a unit disturbance in T_0 is 0.1429. This result is expected because a PD controller has no integral action. The proportional part reduces the disturbance effect, and the derivative part improves the transient response, but neither term can force the final error to zero.

5. CHE Calculator Verification of the Offset

The original PD controller was simulated in the CHE Calculator using the full third-order process model. The process was entered using the following values:

- Process gain = 0.004
- Process lag = 1.0 min
- Valve gain = 1.0, valve lag = 1.25 min
- Sensor gain = 1.0, sensor lag = 1.5 min
- Disturbance gain = 1.0, disturbance lag = 1.0 min
- Controller gain $K_c = 1500$, derivative time $\tau_D = 0.5$ min, no integral time

The set-point change was set to zero and the disturbance load change was set to one. This directly simulated a unit step change in T_0 . The simulated response settled to the same nonzero final value calculated analytically, approximately 0.1429. This confirms that the original PD controller reduces the disturbance response but does not eliminate the offset.

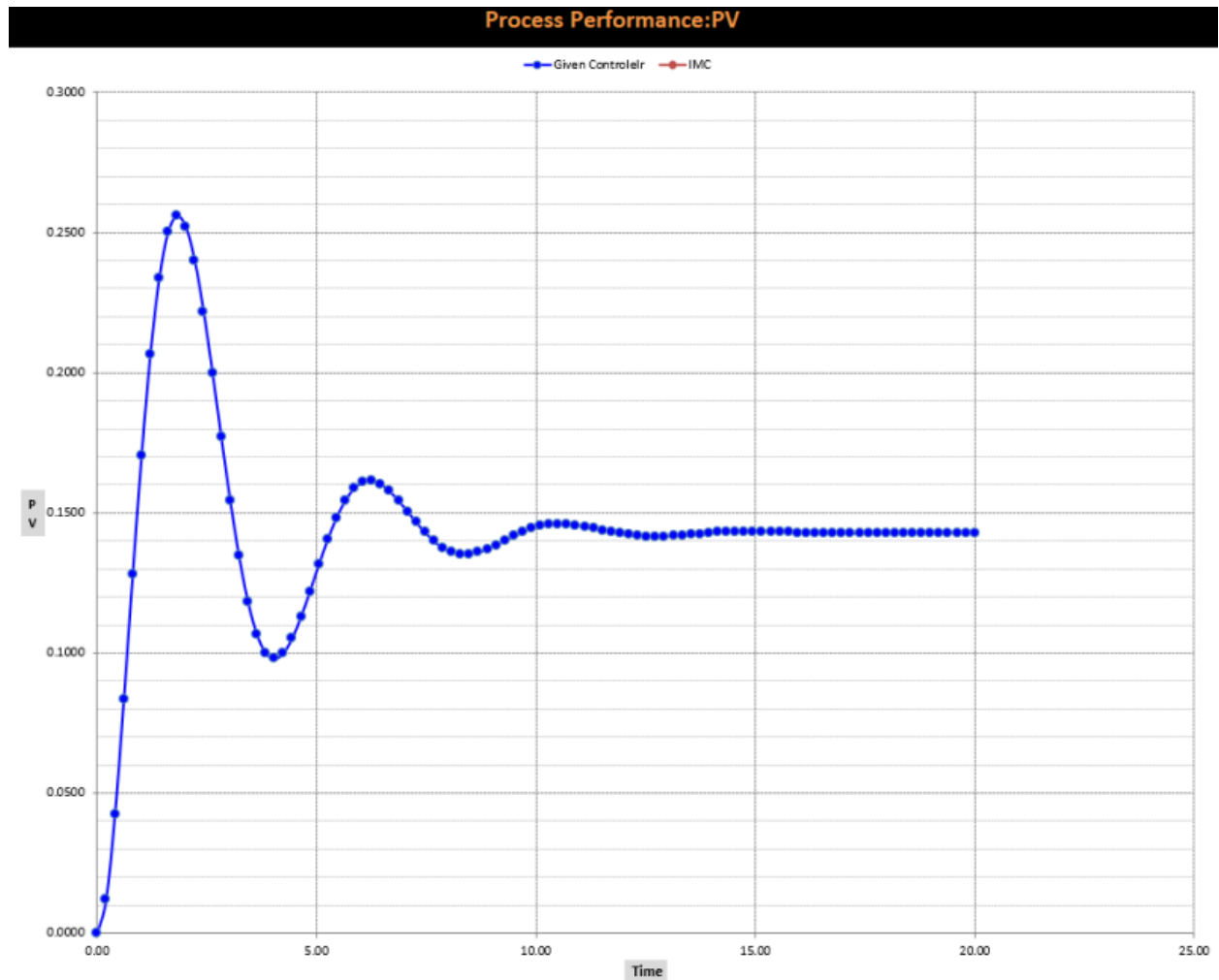


Figure 2: CHE Calculator disturbance response for the original PD controller.

6. Controller Redesign to Eliminate Offset

To eliminate the steady-state offset, integral action must be added. The redesigned controller was therefore selected as a PID controller:

$$G_c(s) = K_c \left(1 + \frac{1}{\tau_I s} + \tau_D s \right)$$

The integral term is the key change because it continues adjusting the heat input until the steady-state error is removed. The same full third-order process was retained for simulation. A reduced model was used only to calculate the controller settings.

The original process model is,

$$G_p(s) = \frac{0.004}{(1.0s + 1)(1.25s + 1)(1.5s + 1)}$$

The third-order model was reduced to a second-order-plus-time-delay model using the half-rule. With time constants ordered as 1.5, 1.25, and 1.0 min, the smallest lag was absorbed into an effective time delay and adjusted time constant:

$$\tau = 1.25 + \frac{1.0}{2} = 1.75 \text{ min}$$

$$\theta^* = 0 + \frac{1.0}{2} = 0.50 \text{ min}$$

The approximate SOPTD model used for IMC tuning was

$$G_p(s) \approx \frac{0.004e^{-0.5s}}{(1.75s + 1)(1.25s + 1)}$$

For the SOPTD IMC PID tuning equations,

$$K_c = \frac{\tau_1 + \tau_2}{K_p(\tau_c + \theta)}$$

$$\tau_I = \tau_1 + \tau_2$$

$$\tau_D = \frac{\tau_1 \tau_2}{\tau_1 + \tau_2}$$

Using $\tau_1 = 1.75 \text{ min}$, $\tau_2 = 1.25 \text{ min}$, $\theta = 0.50 \text{ min}$, and $\tau_c = 0.50 \text{ min}$:

$$\tau_I = 1.75 + 1.25 = 3.00 \text{ min}$$

$$\tau_D = \frac{(1.75)(1.25)}{1.75 + 1.25} = 0.729 \text{ min}$$

$$K_c = \frac{1.75 + 1.25}{0.004(0.50 + 0.50)} = 750$$

The redesigned controller is therefore

$$G_c(s) = 750 \left(1 + \frac{1}{3.000s} + 0.729s \right)$$

When these parameters were entered into the CHE Calculator, the process variable returned to zero after the disturbance. This verified that the addition of integral action eliminated the offset.

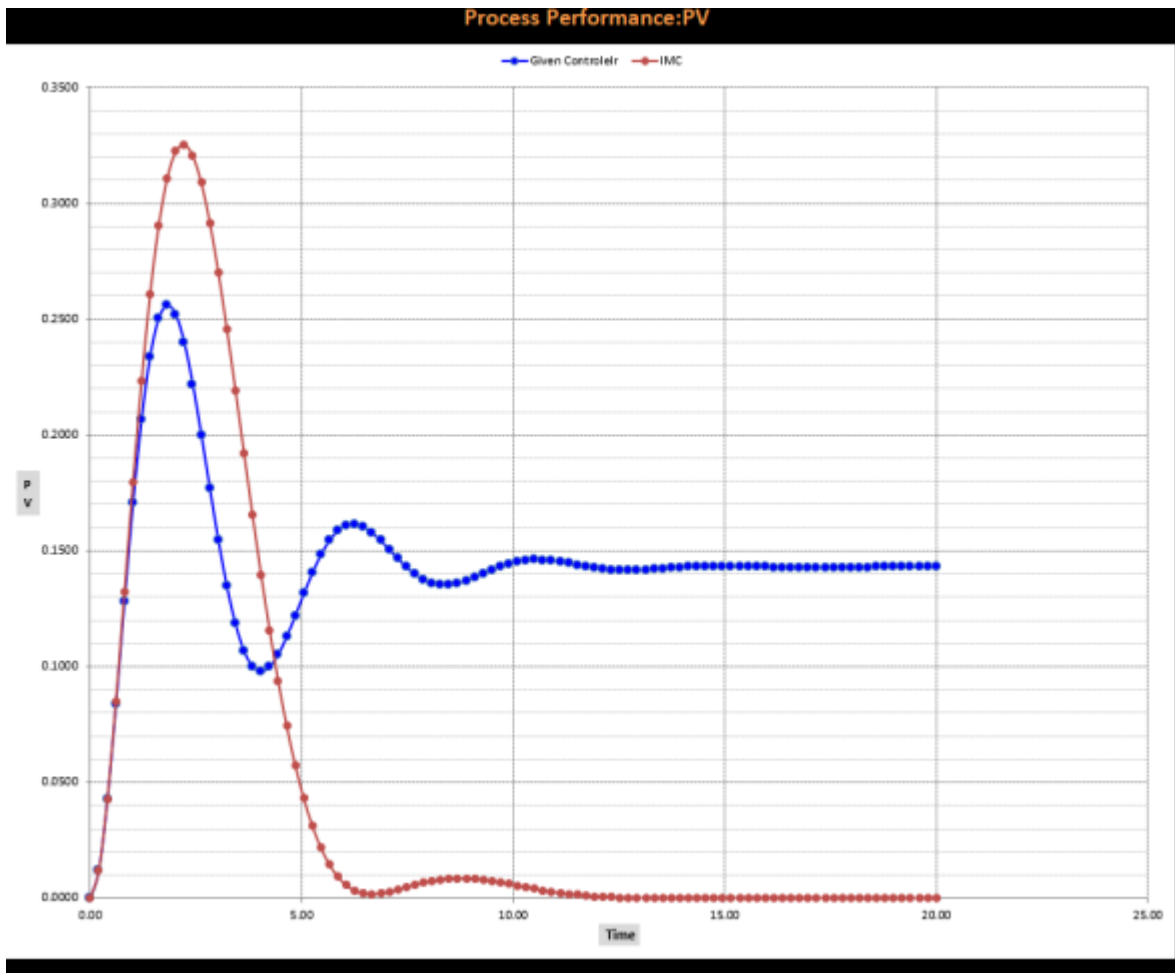


Figure 3: Disturbance-response comparison for the original PD controller and redesigned PID controller.

7. Settling Time and Final Comparison

The controller-performance comparison is summarized in Table 1. The original PD controller has a smaller initial peak, but it settles at a nonzero process-variable deviation. The redesigned PID controller has a larger temporary deviation, but it returns to zero and therefore satisfies the design objective.

Controller	Kc	tau_I (min)	tau_D (min)	Offset	Peak / Overshoot	Settling Time
Original PD	1500	–	0.50	0.1429	0.2560	Does not settle to zero
SOPTD-IMC PID	750	3.00	0.729	0	0.3254	5.6566 min

Overall, the redesigned SOPTD-IMC PID controller is preferred for this system. The original PD controller reduces the disturbance but leaves a steady-state offset. The PID controller introduces integral action, returns the process variable to zero, and reaches a finite settling time of about 5.66 min.

Problem 2: Feedforward Control of a Distillation Column Tray Temperature

Problem Setup

The controlled variable is the critical tray temperature, T . The feed flow rate, F , is the measured disturbance, and the reflux flow rate, L_0 , is the manipulated variable. Separate step tests gave the design transfer functions for the disturbance path and the manipulated-variable path:

$$G_D(s) = \frac{T(s)}{F(s)} = \frac{20e^{-15s}}{30s + 1}$$

$$G_P(s) = \frac{T(s)}{L_0(s)} = \frac{-5e^{-2s}}{6s + 1}$$

The positive sign of G_D means that a feed-rate increase raises the tray temperature. The negative sign of G_P means that increasing reflux lowers the tray temperature. The actual plant used for simulation differs from the design model:

$$T(s) = \left[\frac{-4e^{-2s}}{6s + 1} \right] L_0(s) + \left[\frac{25e^{-15s}}{30s + 1} \right] F(s)$$

This model mismatch is important because feedforward control can reject disturbances perfectly only when the process and disturbance models are accurate.

(a) Feedforward Controller Design and Realizability

For perfect disturbance rejection, the feedforward controller must cancel the disturbance contribution to the output. The ideal feedforward relation is

$$G_{FF}(s) = -\frac{G_D(s)}{G_P(s)}$$

Substituting the design models gives

$$G_{FF}(s) = -\frac{\frac{20e^{-15s}}{30s + 1}}{\frac{-5e^{-2s}}{6s + 1}}$$

$$G_{FF}(s) = 4e^{-13s} \frac{6s + 1}{30s + 1}$$

Therefore, the feedforward controller has a gain of 4, a lead time of 6 min, a lag time of 30 min, and a time delay of 13 min.

The controller is realizable because the disturbance path has a larger delay than the process path. The feedforward delay is

$$\theta_{FF} = \theta_D - \theta_P = 15 - 2 = 13 \text{ min}$$

Since this delay is positive, the controller does not require future knowledge of the disturbance. The lead-lag portion is also proper because the denominator lag is larger than the numerator lead.

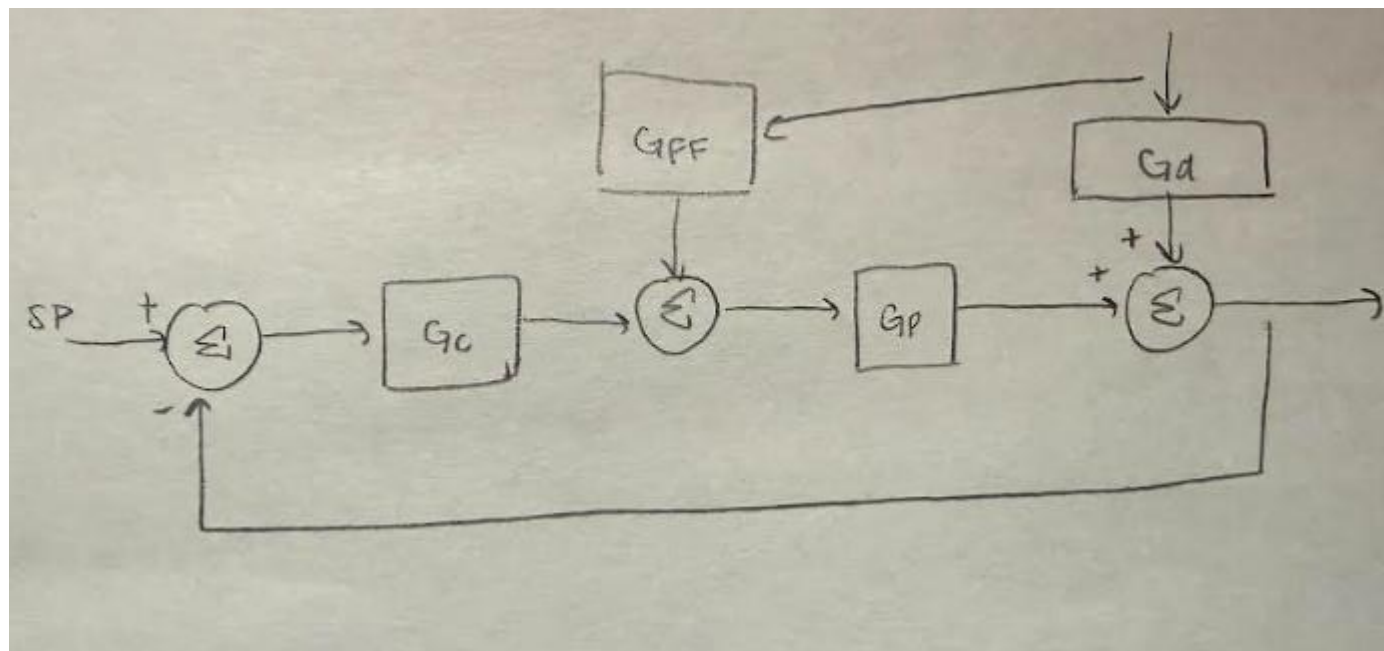


Figure 4: Feedforward-feedback block diagram for the distillation tray-temperature control loop.

(b) Feedforward-Only Control Applied to the Actual Process

The feedforward controller from part (a) was applied to the actual process. In the CHE Calculator, the manipulated-variable model was entered with a gain of -4, a 6-minute lag, and a 2-minute delay. The actual disturbance model was entered with a gain of 25, a lag of 30 minutes, and a delay of 15 minutes. The feedforward controller was entered with a gain of 4, a delay of 13 minutes, a lead of 6 minutes, and a lag of 30 min.

A feed disturbance step of $DS = 2$ was then applied while feedback control was not active. The response did not return to zero. Instead, it settled at a large positive offset. This happens because the feedforward controller was designed based on the assumed model, but the actual plant has a larger disturbance gain and a smaller manipulated-variable gain.

The final value can be checked using steady-state gains. For a disturbance step of 2, the direct disturbance effect is

$$25(2) = 50$$

The feedforward controller produces a steady-state control action of

$$4(2) = 8$$

Passing this through the actual process gain gives

$$(-4)(8) = -32$$

Therefore, the net steady-state temperature deviation is

$$50 - 32 = 18$$

This agrees with the feedforward-only response. The controller works in principle, but model mismatch prevents complete disturbance cancellation.

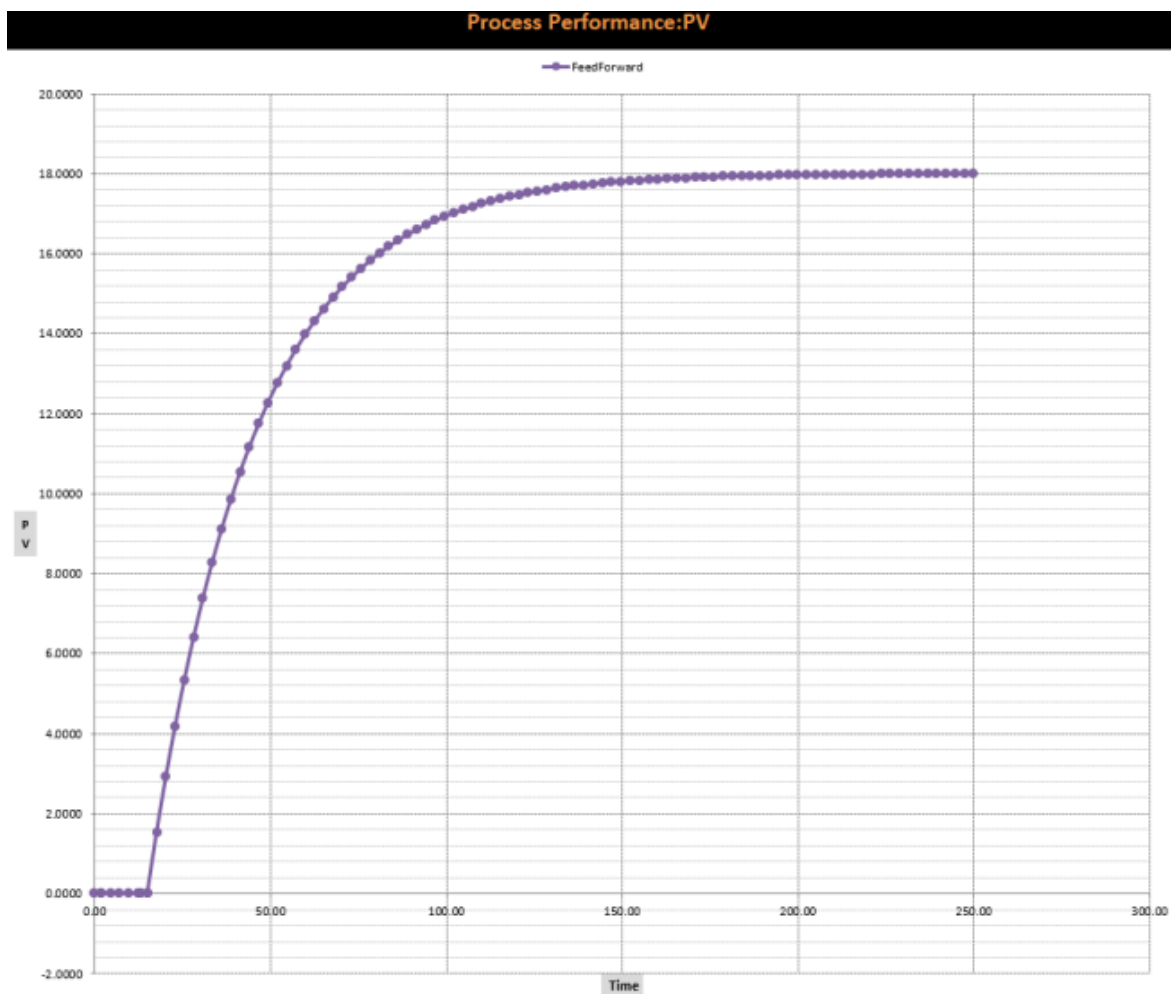


Figure 5: Feedforward-only response of the actual process to $DS = 2$.

(c) Feedback Controller Design by Direct Synthesis

A feedback controller was designed using the assumed process model from Equation (1), as required. For a FOPTD process model,

$$G_p(s) = \frac{K_p e^{-\theta s}}{\tau s + 1}$$

the direct-synthesis PI tuning relations are

$$K_c = \frac{\tau}{K_p(\lambda + \theta)}, \quad \tau_I = \tau$$

Using the assumed process model

$$G_p(s) = \frac{-5e^{-2s}}{6s + 1}$$

the parameters are $K_p = -5$, $\tau = 6$ min, $\theta = 2$ min, and $\lambda = 0.25$ min. Substitution gives

$$K_c = \frac{6}{(-5)(0.25 + 2)} = -0.5333$$
$$\tau_I = 6 \text{ min}$$

The feedback controller is therefore

$$G_c(s) = -0.5333 \left(1 + \frac{1}{6s} \right)$$

This controller was applied to the actual process without feedforward action. Because it contains integral action, the final offset is removed. However, feedback cannot act until the disturbance has already affected the tray temperature, so the response shows a larger peak and a slower return to zero than the combined feedforward-feedback case.

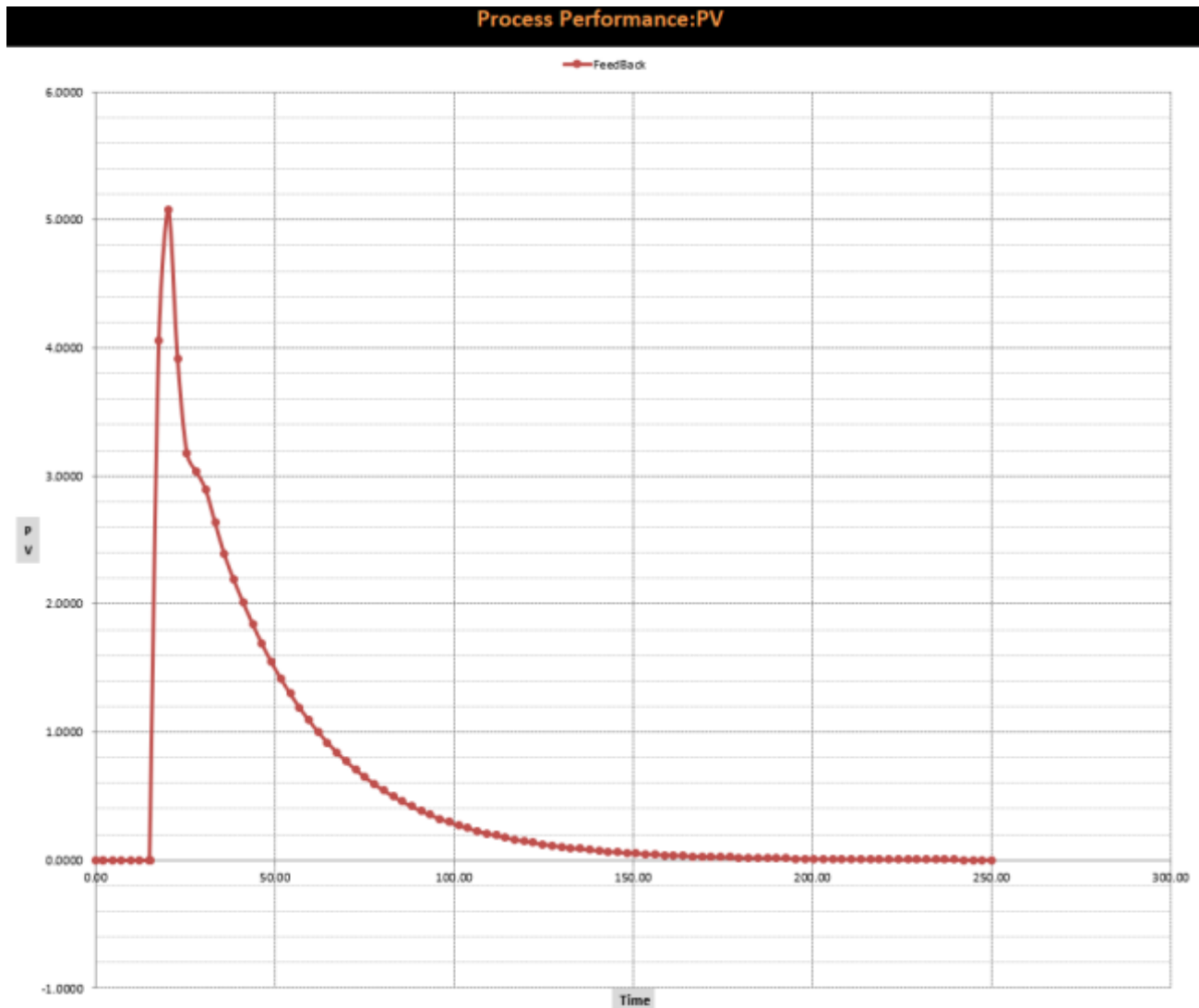


Figure 6: Feedback-only response of the actual process to $DS = 2$.

(d) Combined Feedforward-Feedback Control

The final simulation used both the feedforward controller from part (a) and the feedback controller from part (c). The process and disturbance models were the same actual models used in the previous simulations.

This control structure combines the strengths of both methods. The feedforward controller responds as soon as the measured feed disturbance changes, reducing the initial effect of the upset. The feedback controller then removes the remaining error caused by model mismatch. This is useful here because the feedforward controller alone cannot perfectly cancel the actual disturbance.

The combined response returns to zero and has a smaller peak than the feedback-only case. Compared with feedforward-only control, it also removes the final offset. This confirms that the combined strategy gives the best overall disturbance rejection for the mismatched actual process.

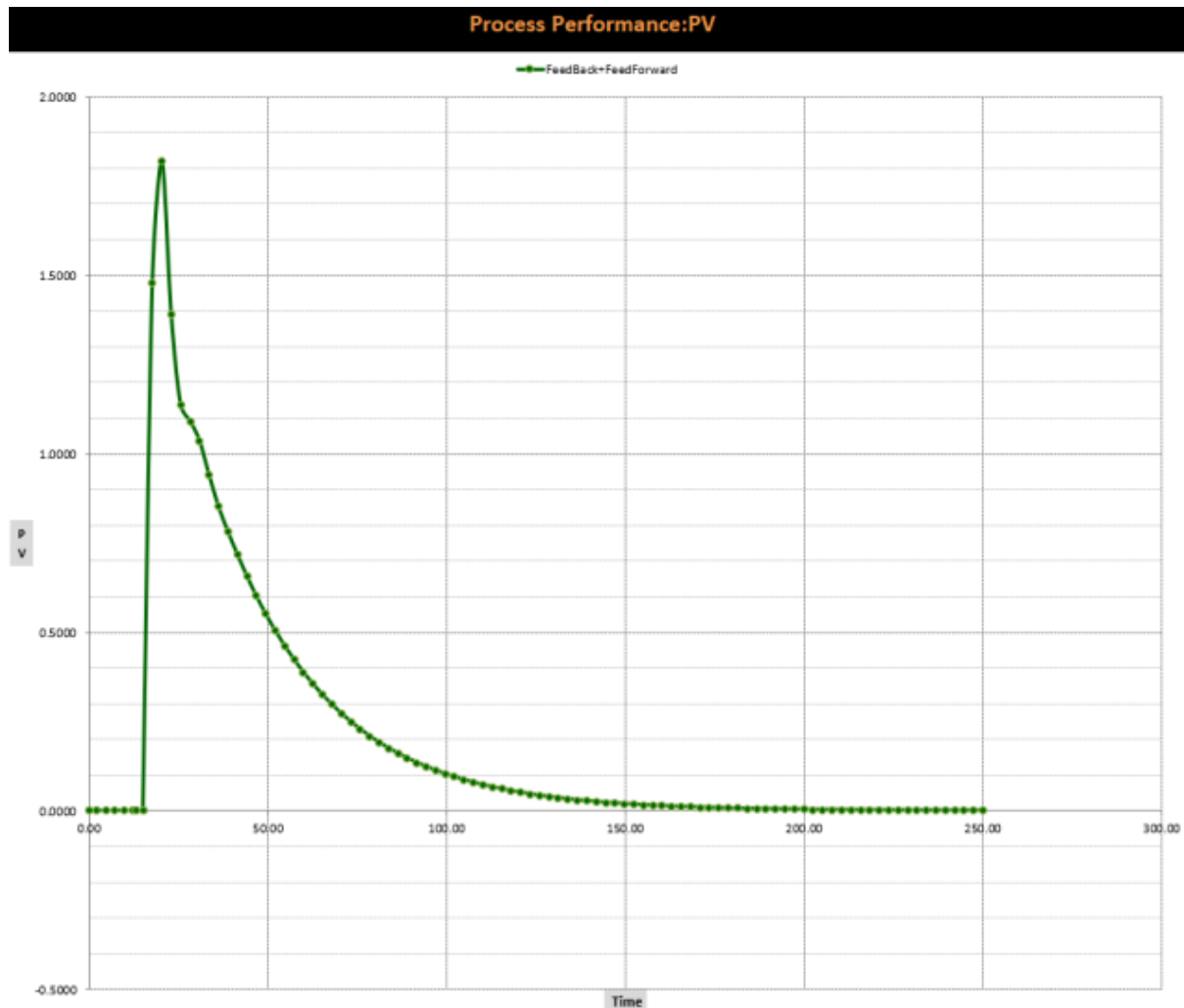


Figure 7: Combined feedforward-feedback response of the actual process to $DS = 2$.

(e) Comparison of the Three Responses

All three response curves were plotted on the same graph for the same disturbance step, $DS = 2$. The feedforward-only response acted before the tray temperature changed, which is expected because it uses the measured disturbance directly. However, it did not return to zero and instead settled near 18. This occurred because the feedforward controller was based on the nominal model, while the actual process gains were different.

The feedback-only response removed the final offset because the controller contains integral action. However, the response had the largest peak and the slowest settling behavior because feedback control reacts only after the disturbance has affected the process variable.

The combined feedforward-feedback response performed best. Feedforward reduced the initial disturbance effect, and feedback removed the residual error caused by imperfect model cancellation. Therefore, the feedforward controller did not perform perfectly by itself on the actual process, but it did perform as expected once model mismatch is considered.

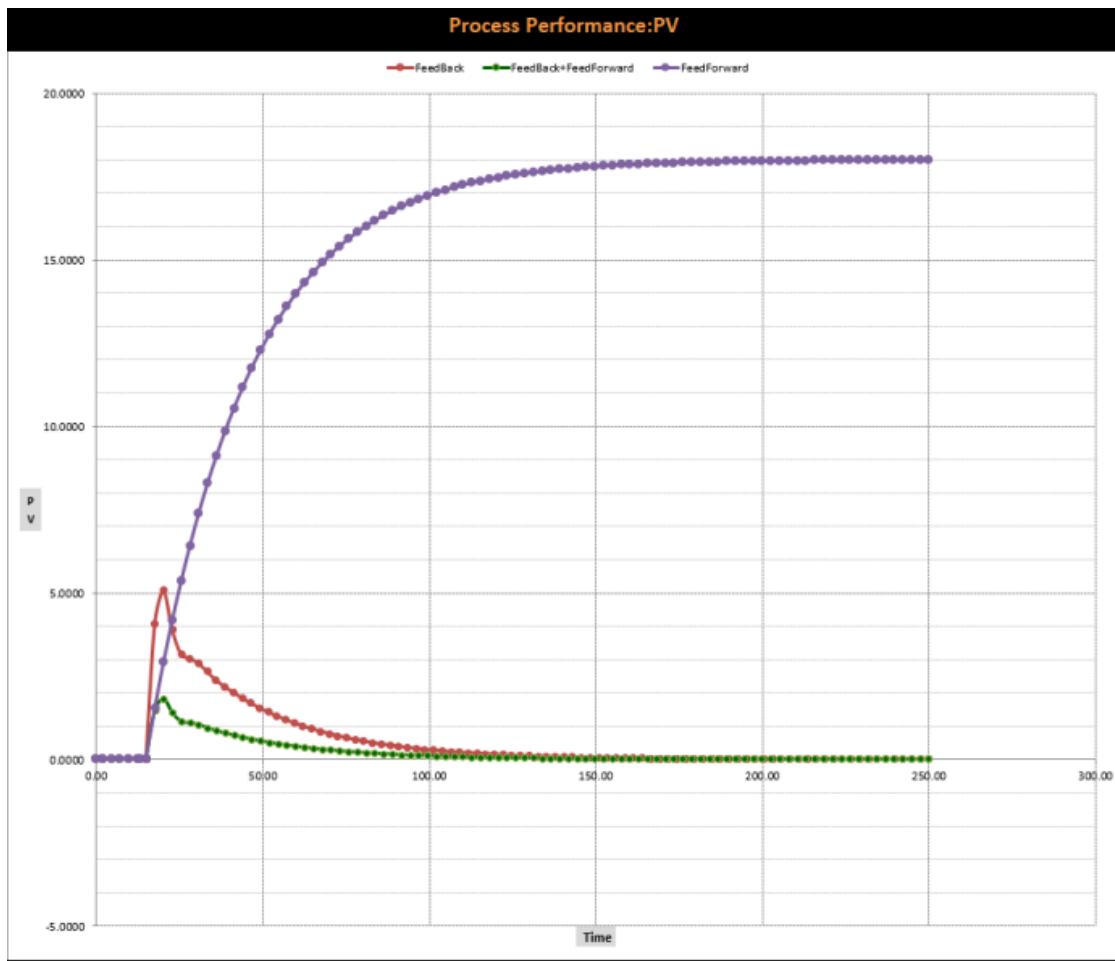


Figure 8: Combined plot of feedforward-only, feedback-only, and feedforward-feedback responses.

Controller	Offset	Peak / Overshoot	Settling Time
Feedforward only	18.0	0	Does not settle to zero
Feedback only	0	5.083	179.5 min
Feedforward + Feedback	0	2.91	149.66 min

The table confirms the trends seen in the plot. Feedforward-only control has the largest steady-state error because the model cancellation is not exact. Feedback-only control eliminates the offset but responds slowly. The combined strategy gives the most useful result because it lowers the peak compared with feedback alone while still eliminating offset.

References and Use Statement

Caracotsios, Michael. *CHE Calculator*, Version 13. Excel-based chemical engineering calculation and dynamic modeling tool. University of Illinois Chicago, Department of Chemical Engineering, 2026.

Caracotsios, Michael. *Introduction to Chemical Process Control*, Chapters 5-7. University of Illinois Chicago, Department of Chemical Engineering, 2026.

OpenAI. ChatGPT. Used to help revise wording, organize the report, improve clarity, and check consistency of the written explanation. The calculations, simulations, and final engineering conclusions were reviewed and selected by the student.